

JEE MAIN REVISION MASTER: PROPERTIES OF TRIANGLES

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 4 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Properties Of Triangle — Standard Calculations & Problem Archetypes	3 PYQs	Quantitative Analysis
Standard	Properties Of Triangle — Core Principles & Governing Laws	1 PYQs	Core Principle
Standard	Properties Of Triangle — Applied Cases, Reactions & Traps	0 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Properties Of Triangle — Standard Calculations & Problem Archetypes	$(x_1, y_1), \quad (x_2, y_2)$
Properties Of Triangle — Core Principles & Governing Laws	$(a-d)+a+(a+d)=180^\circ, \quad 180^\circ$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Properties Of Triangle — Standard Calculations & Problem Archetypes (Moderate Yield — 3 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Properties Of Triangle.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2024 Shift] Q#2 (MCQ): "If $\triangle ABC$ is right angled at C , then the value of $\tan A + \tan B$ is..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "The area of a triangle with vertices $(-3, 0)$, $(3, 0)$ and $(0, k)$ is 9 sq units, find the value of k is..." [Ans: N/A]
- [2017 Shift] Q#4 (MCQ): "The area of triangle with vertices $(K, 0)$, $(4, 0)$, $(0, 2)$ is 4 sq units, then value of K is..." [Ans: N/A]

3.2 Properties Of Triangle — Core Principles & Governing Laws (Standard Yield — 1 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Properties Of Triangle.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Properties Of Triangle.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Properties Of Triangle PYQs.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "The angles of a triangle are in A.P and the greatest angle is double the least angle, then sine of the third angle is..." [Ans: A]

3.3 Properties Of Triangle — Applied Cases, Reactions & Traps (Standard Yield — 0 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Properties Of Triangle.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "The angles of a triangle are in A.P and the greatest angle is double the least angle, then sine of the third angle is..." [Ans: A]
- **[2024 Shift] Q#2 (MCQ):** "If $\triangle ABC$ is right angled at C , then the value of $\tan A + \tan B$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The angles of a triangle are in A.P and the greatest angle is double the least angle, then sine of the third angle is

- (A) $\frac{\sqrt{3}}{2}$ [Correct]
- (B) $\frac{1}{2}$
- (C) $\frac{1}{\sqrt{2}}$
- (D) 0

Correct Answer: A

Solution Analysis: Let the three angles of the triangle be in A.P. So we can take them as $a-d, a, a+d$. Since the sum of angles of a triangle is 180° , $(a-d) + a + (a+d) = 180^\circ$ $3a = 180^\circ$ $a = 60^\circ$. So the middle angle is 60° . Now, the greatest angle is double the least angle. $\text{Greatest angle} = a+d$ $\text{Least angle} = a-d$ So, $a+d = 2(a-d)$ Putting $a = 60^\circ$, $60^\circ + d = 2(60^\circ - d)$ $60^\circ + d = 120^\circ - 2d$ $3d = 60^\circ$ $d = 20^\circ$. Thus the angles are $40^\circ, 60^\circ, 80^\circ$. The third angle in the A.P. is the middle angle, 60° . So, $\sin 60^\circ = \frac{\sqrt{3}}{2}$

Benchmark Problem 2: KCET 2024 (Q#2)

If $\triangle ABC$ is right angled at C , then the value of $\tan A + \tan B$ is

- (A) $a+b$
- (B) a^2/bc
- (C) c^2/ab
- (D) b^2/ac

Correct Answer: None

Solution Analysis: $\tan A = (a/b), \tan B = (b/a) \therefore \tan A + \tan B = (a/b) + (b/a) = \frac{a^2 + b^2}{ab} = \frac{c^2}{ab}$ [$c^2 = a^2 + b^2$]